

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 5: Adapted Coordinates and Uniqueness

Munhyeok Kang

2026-09-28–2026-10-04

Overview

Adapted coordinate conditions are formulated as identities among field-equation coefficients rather than as a coordinate choice made after solving.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$B_\mu(g, \partial g) = 0$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$T_a[g(x)] = T_a[g'(x')](a \in \text{Registry})$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$\mathcal{E}_{C5'}(c_1, \dots, c_n) = 0$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Einstein–Mythos paper B, §§2–7
- Hall, ch. 5
- Adapted Coordinate Standard ACS-12